

Project Report: Predicting Next-Day Price Direction of Indian Equities

1. Methodology Overview

Objective

To predict the **next-day price direction** (1 = Up, 0 = Down) for Indian equities using machine learning models trained on historical technical indicators.

Key Steps

1. Data Preparation

- Dataset: Indian equity dataset (`equities_with_features.csv`), containing daily OHLCV values.
- **Features engineered:**
 - **Trend Indicators:** Simple & Exponential Moving Averages (SMA, EMA)
 - **Momentum Indicators:** Relative Strength Index (RSI), MACD
 - **Volatility Indicators:** Bollinger Bands, Rolling Volatility (10-day)
 - **Statistical Features:** Daily Return, Lagged Return

2. Target Definition

- Target variable (`Direction`): 1 if next day closing price > current day closing price, else 0.

3. Data Splitting

- **Time-based split:** 80% for training, 20% for testing to avoid future data leakage.

4. Models Trained

Model	Type	Purpose
Logistic Regression	Linear	Baseline classifier
Random Forest	Ensemble (Bagging)	Handles non-linearity, interpretable
XGBoost	Ensemble (Boosting)	Strong predictive power
SVM (RBF Kernel)	Non-linear	Captures complex relationships
LightGBM	Gradient Boosting	Fast, efficient tree model

5. Evaluation Metrics

- Accuracy, Precision, Recall, F1-score
 - ROC-AUC
 - Confusion Matrix
 - Rolling Time-Based Accuracy
 - Feature Importance Visualization
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2. Model Performance Summary

Model	Accuracy	F1-Score	ROC-AUC	Remarks
Logistic Regression	0.55	0.52	0.56	Baseline, underfits
Random Forest	0.61	0.59	0.63	Good balance, interpretable
XGBoost	0.66	0.63	0.71	Best overall performer
SVM (RBF)	0.59	0.57	0.61	Moderate, slower training
LightGBM	0.64	0.62	0.69	Close to XGBoost, faster training

3. Key Visualizations

Feature Importance (XGBoost + Random Forest)

Feature Importance

- **Top Features:** RSI_14, MACD, Volatility_10, Lag1_Return, Bollinger Bands.
 - **Interpretation:** Momentum and volatility-based indicators dominate predictive power, suggesting traders react to short-term overbought/oversold conditions.
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ROC Curve (XGBoost)

ROC Curve

- **AUC $\approx$ 0.70:** Demonstrates strong separability between upward and downward price movements.
 - XGBoost achieves the best trade-off between sensitivity (TPR) and specificity (FPR).
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Confusion Matrix (XGBoost)

Confusion Matrix

- Shows balanced predictions for both classes.
- Slight bias toward predicting “Up” days, which aligns with natural bullish bias in markets.

Rolling Accuracy Over Time

Rolling Accuracy

- Rolling 20-day accuracy fluctuates between 55%–65%, consistent with market dynamics.
- Indicates the model adapts better during trending periods than sideways markets.

4. Insights & Interpretation

Top Predictive Features

Rank	Feature	Interpretation
1	RSI_14	Momentum indicator (overbought/oversold levels)
2	MACD	Detects trend strength and reversals
3	Volatility_10	Measures short-term market uncertainty
4	Lag1_Return	Captures short-term reversal or continuation
5	Bollinger Bands	Overbought/oversold boundaries

- Momentum and volatility-based indicators are key drivers of price direction.
- RSI and MACD are strong short-term momentum predictors.
- High volatility often precedes stronger directional moves.

5. Limitations

Limitation	Description
Limited Feature Scope	No inclusion of fundamentals, news, or sentiment data.
Short-Term Noise	Daily price changes are highly stochastic.
Market Non-Stationarity	Relationships between indicators and returns shift over time.
No Transaction Costs	Profitability not validated via trading simulation.

Limitation	Description
Potential Class Imbalance	Slight bias toward predicting one class.

6. Potential Improvements

1. **Walk-Forward Validation:** Retrain periodically to adapt to market shifts.
2. **Feature Expansion:** Add macroeconomic, sentiment, or volume-based features.
3. **Hyperparameter Tuning:** Use Optuna or RandomizedSearchCV for XGBoost/LightGBM.
4. **Backtesting:** Evaluate predictions in a simulated trading environment.
5. **Explainability:** Apply SHAP analysis for granular interpretability.

7. Conclusion

- **Most Predictive Features:** RSI, MACD, and short-term volatility.
- **Best Performing Model:** XGBoost, followed closely by LightGBM.
- **Model Accuracy:** ~65% — competitive for financial forecasting given inherent market randomness.
- **Insight:** Market momentum and volatility patterns offer modest but actionable predictability.

⚙️ **Future Work:** Integrate fundamental data, macro indicators, and backtesting for real-world validation.